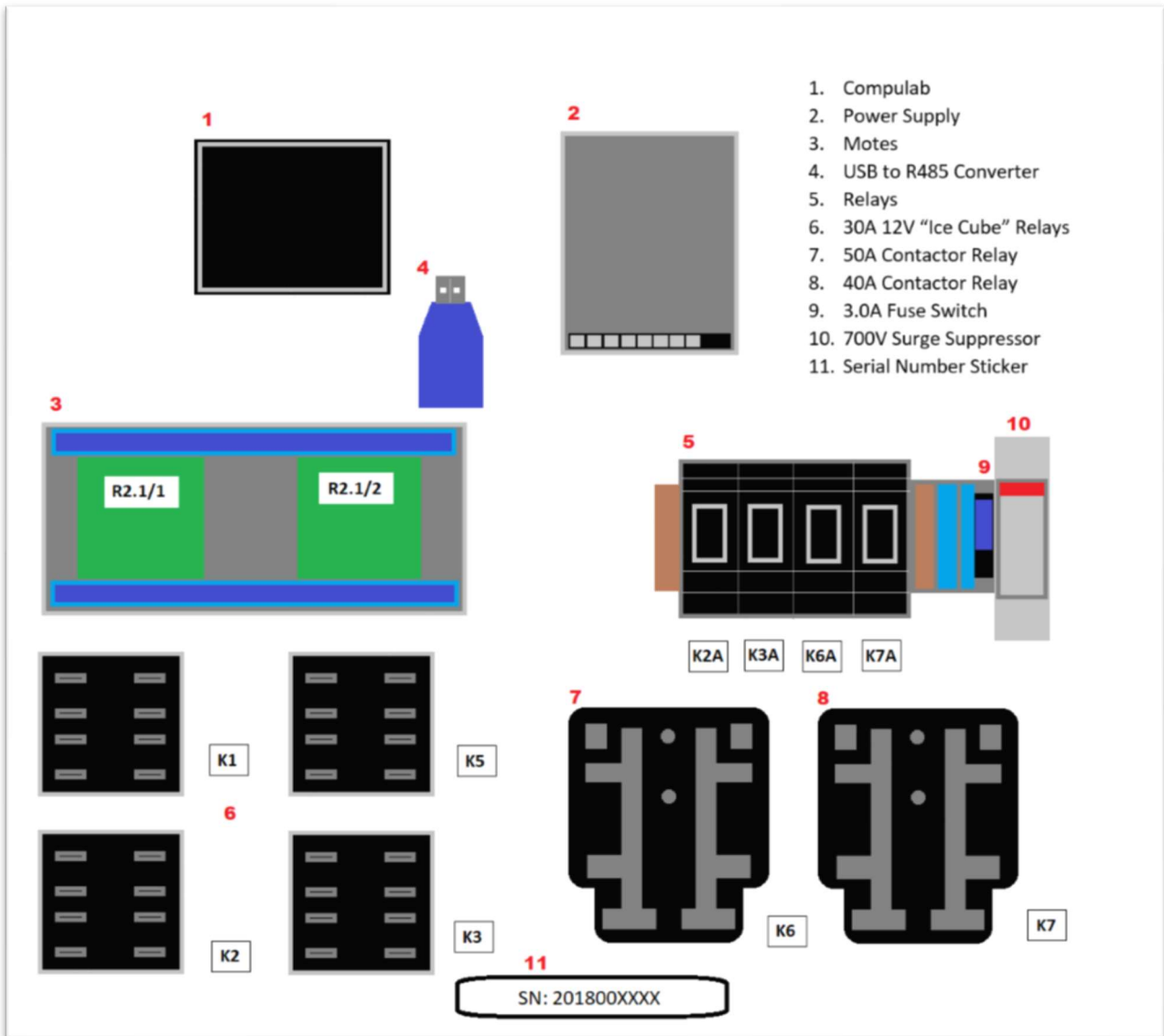


Residential System Manual

The Dencor Residential Energy Management System (EMS) is comprised of the Compulab, Motes, Power Supply, Relays, and Contactors. The system monitors both grid and solar output through Current Transformers to micromanage power to prominent loads for a residential location. The system works autonomously once the settings are in place, but a native 4G network built into the Compulab let's Dencor technicians service the unit, and perform updates, remotely.



The diagram above details the components found in the enclosure. The EMS enclosure is typically mounted outside near the main electrical panels for easy access to the loads, and main junction box. After the system is installed, only qualified personnel should access the box, for both safety and to maintain the proper functionality of the system.

The Compulab is the computer that processes data and sheds load to lower electrical energy usage around the clock. The settings in the system are changed to suit utility structures in the resident's region. The system has an embedded portal accessible through a local connection, or WiFi. This webpage shows the loads, their current shed status, as well as CT readings.



A Dencor Technician will setup this page as well as shed priority, and KW setpoint at the time of the install. By default, the address to connect to the local system through a webpage is 10.1.0.100.

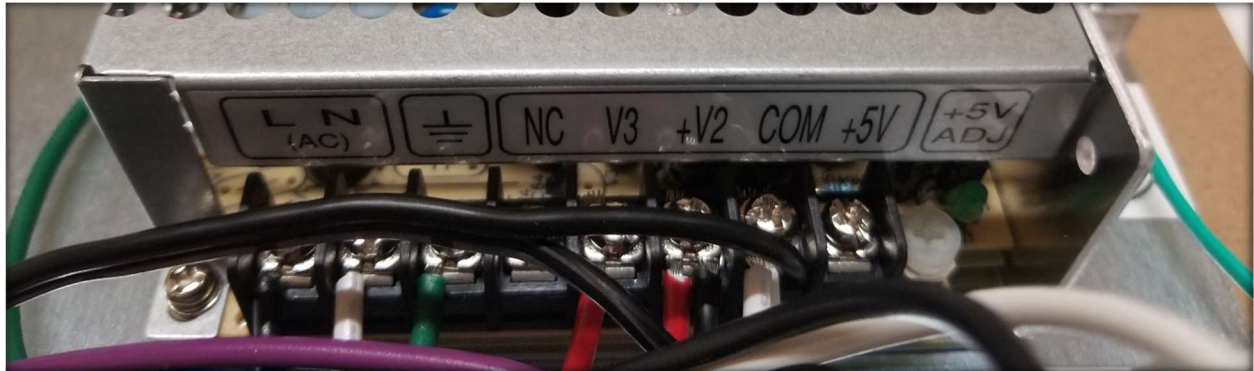
The Compulab is powered via 24VDC from the Power Supply. It also features an antenna mounted on the outside of the enclosure for the native 4G connection. The antenna is plugged into the top left port, with a male-female adaptor. The bottom left port is to improve WiFi connection.

The blue **USB-485 converter** is plugged into the second USB port from the bottom right and allows the Compulab to communicate with the Motes.

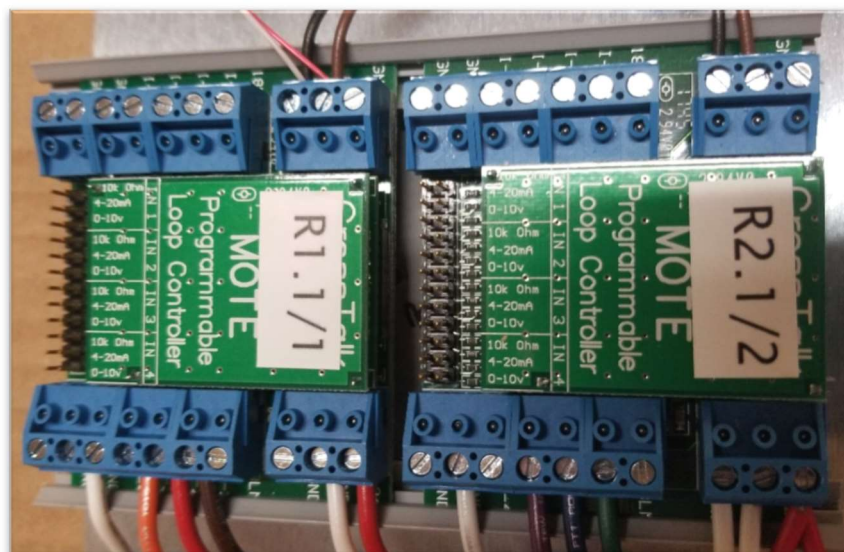
There are two **ethernet ports**. Facing the enclosure, the top ethernet port is for accessing 10.1.0.100, while the left-sided port is an alternate port for 192.168.14.100. Only the top port is used during setup.

The bottom left edge has a power button, but the **3.0A Fuse Switch** is the recommended way to power on/off.

The **Power Supply** provides different voltages to the whole assembly, primarily 12VDC and 24VDC. The Voltages are labeled above the terminals as shown below, and the unit features a green LED indicating power. The **Compulab** and **Motes** run on 24VDC, while the **Relay** circuit uses 12VDC,



The **Mote** is a micro-controller that receives and sends information with the Compulab. It features four inputs and four outputs. In the standard residential system, there are two motes.



The **Motes** are designated by a version number and physical position. In the above image, the mote on the left is version R1.1 (R for Residential) and /1 for the first position in the layout.

The first mote's **inputs** are used for the grid power output CTs, while mote two is for solar power output CTs. Using jumpers, the input type can be changed between 0-10V, 4-20mA, and 10K Ohms. The standard system uses 0-10V CTs.

The 8 **outputs** between the motes are what send signals to the **relays** to turn devices on/off. In Residential systems, only six outputs are used in conjunction with relays to accommodate different load sizes. The diagram below shows how the layout is wired to cover the ranges.

